

Yao Ming (Brian) Chan

PERSONAL INFORMATION

Email: brichan17@gmail.com

Date of birth: November 17th 1999

Place of birth: Ipoh, Malaysia

Nationality: Australian

Website: <https://brichan17.github.io>

EDUCATION

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|-----------------|---|
| 10/2023–Now | DPhil Mathematics
University of Oxford
Expected graduation date: July 2027
Current thesis title: On the Cuntz semigroups of non-simple C^* -algebras
Supervisor: Professor Stuart White |
| 03/2021–11/2022 | MSc. Master of Science (Mathematics and Statistics)
University of Melbourne
Major: Pure Mathematics
Thesis: Wedge product matrices and applications
Supervisor: Professor Arun Ram |
| 03/2018–11/2020 | BSc. Bachelor of Science
University of Melbourne
Major: Mathematics and Statistics
Specialisation: Pure Mathematics |

TEACHING

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| 2025 | Class tutor for Functional Analysis I B4.1

Organised and taught one set of classes (four classes total). Also played the role of the teaching assistant and marked the students' submitted work before each class. |
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2025	Teaching assistant for Functional Analysis II B4.2
	Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class.
2023, 2024	Teaching assistant for Functional Analysis I B4.1
	Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class.

ACHIEVEMENTS

2021, 2022	Mathematics and Statistics Masters Scholarship
	Three payments of \$2000 corresponding to the first three semesters of the Master of Science degree in recognition of consistent academic achievement throughout the degree.
2018, 2019	Dean's Honours List
	Awarded for an average mark which lies in the top 3% of students in the Bachelor of Science degree.

PREPRINTS AND PUBLICATIONS

1. Y. Chan. *Wedge product matrices and orbits of principal congruence subgroups*, Lin Alg App. **696** (2024), 1–28.
2. Y. Chan. *The Cuntz semigroup of a unital graph C^* -algebra*, preprint (arXiv: 2610.01915 [math.OA]), 2026.
3. Y. Chan, J. Curda, A. Shiner. *Classification of certain C^* -algebras with a unique ideal*, currently in preparation.

TALKS DELIVERED

29/07/2026	The Cuntz semigroup of a unital graph C^*-algebra (50 minute invited talk), UNSW, Sydney, Australia.
25/06/2026	The Cuntz semigroup of a unital graph C^*-algebra (25 minute contributed talk), UK Operator Algebras Conference, Sabhal Mòr Ostaig, Isle

of Skye.

- 08/06/2026 **The Cuntz semigroup of a unital graph C^* -algebra** (50 minute invited talk), Junior Functional Analysis Seminar, University of Oxford.
- 20/03/2026 **The Cuntz-Krieger relations in the context of the Cuntz semigroup** (15 minute contributed talk), Operator algebras in the south, Cardiff University.
- 11/09/2025 **Purely large extensions with purely infinite quotients** (20 minute contributed talk), Topological phenomena in the Cuntz semigroup, University of Oxford.
- 21/07/2025 **Cuntz semigroups and graph C^* -algebras** (lightning talk), YMC*A, University of Southern Denmark, Odense.
- 04/07/2025 **Cuntz semigroups and graph C^* -algebras** (lightning talk), UK Operator Algebras Conference, Queen's University Belfast.
- 03/04/2025 **Nuclear dimension, extensions and graph C^* -algebras** (20 minute contributed talk), Young Functional Analysts' Workshop, University of Glasgow.
- 12/06/2024 **Nuclear dimension and graph C^* -algebras** (lightning talk), UK Operator Algebras Conference, Newcastle University, Newcastle upon Tyne.

RESEARCH VISITS AND CONFERENCES

- 31/08/2026 - 04/09/2026 **Cuntz semigroups and dynamics** Chalmers University of Technology, Gothenburg, Sweden.
- 29/07/2026 - 31/07/2026 **Research visit** UNSW, Sydney, Australia.
- 21/06/2026 - 26/06/2026 **UK Operator Algebras Conference**, Sabhal Mòr Ostaig, Isle of Skye, Scotland.

- 19/03/2026 - 20/03/2026 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.
- 10/09/2025 - 12/09/2025 **Topological phenomena in the Cuntz semigroup**, University of Oxford, Oxford, United Kingdom.
- 21/07/2025 - 25/07/2025 **Young Mathematicians in C*-algebras**, University of Southern Denmark, Odense, Denmark.
- 02/07/2025 - 04/07/2025 **UK Operator Algebras Conference**, Queen's University Belfast, Belfast, Northern Ireland.
- 02/04/2025 - 04/04/2025 **Young Functional Analysts' Workshop**, University of Glasgow, Glasgow, Scotland.
- 07/03/2025 - 07/03/2025 **Operator algebras in the south**, Cardiff University, Cardiff, Wales.
- 12/06/2024 - 14/06/2024 **UK Operator Algebras Conference**, Newcastle University, Newcastle upon Tyne, United Kingdom.
- 06/11/2023 - 10/11/2023 **Twinned Conference on C*-algebras and Tensor Categories**, The University of Edinburgh, Edinburgh, Scotland.

SKILLS

Coding experience with Python

2021

Quantum chemistry project for the subject COMP90072

Used NumPy to implement the Hückel method in order to model the energy levels, electron densities and bond orders of conjugated organic molecules. The Hartree-Fock method was also implemented to compute the orbital energies of small atoms with occupied 1s and 2s orbitals and diatomic molecules comprised of these atoms.

LANGUAGES

English (Native)

PROFESSIONAL INVOLVEMENT

2025-2026 **Co-organiser** for the “Young Mathematicians in C^* -algebras” conference, held at the University of Oxford in 2026.

Managed the email inbox for the conference. Responded to various queries and requests from participants.

REFERENCES

Professor Arun Ram (MSc supervisor)

Institute: University of Melbourne

Email: aram@unimelb.edu.au

Professor Stuart White (DPhil supervisor)

Institute: University of Oxford

Email: stuart.white@maths.ox.ac.uk